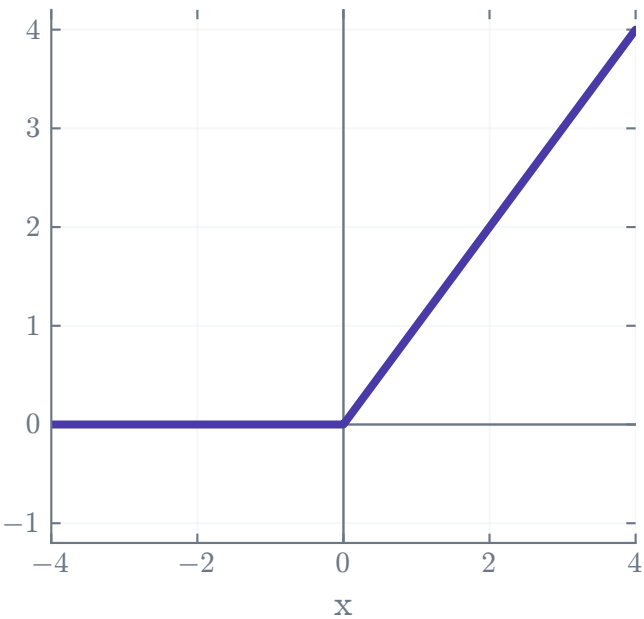


ReLU

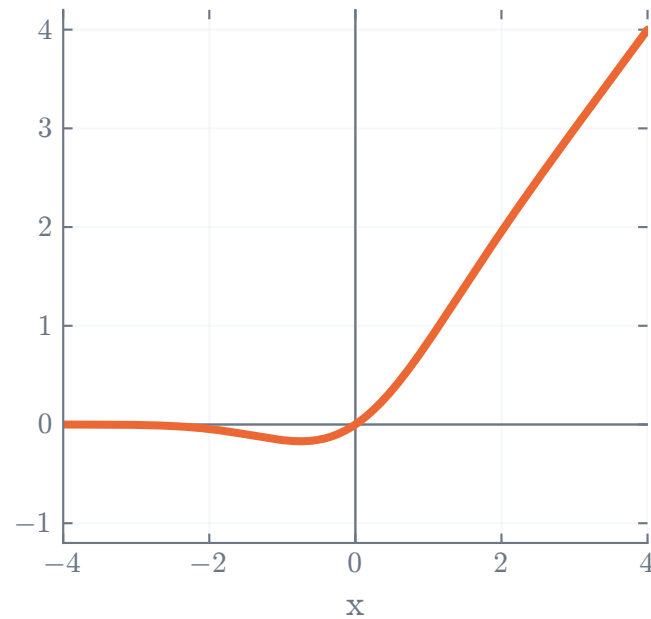
$$\max(0, x)$$



NN

GELU

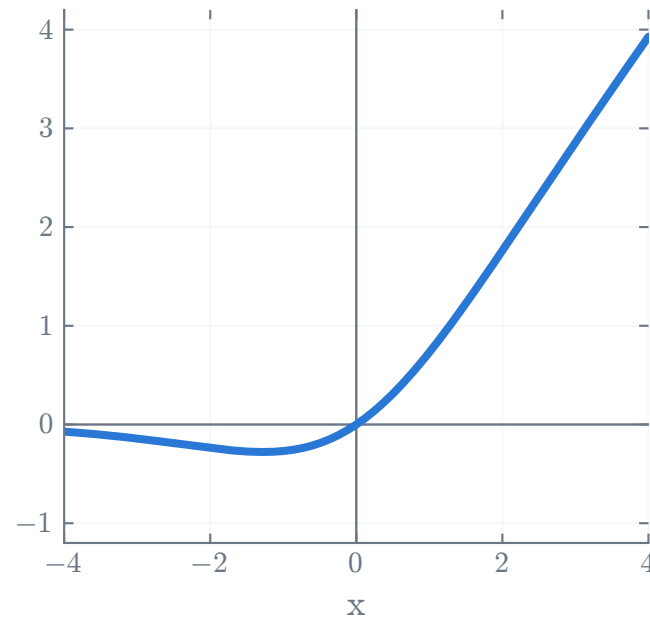
$$x \Phi(x)$$



FNO, WNO, STO

SiLU

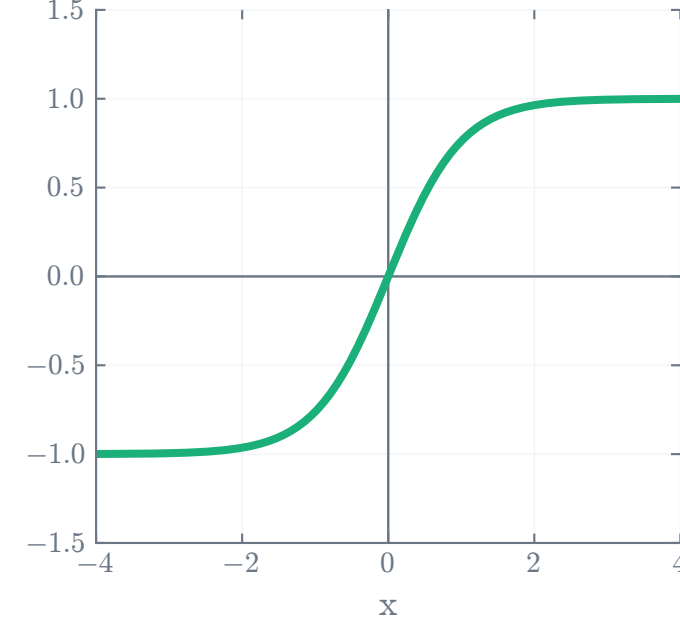
$$x / (1 + e^{-x})$$



DNO, DCO, LNO, GNO, refinement conv

Tanh

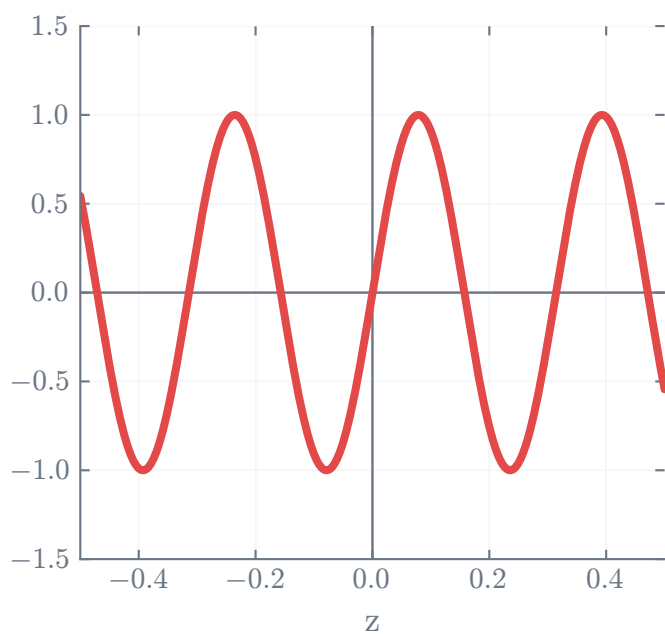
$$(e^x - e^{-x}) / (e^x + e^{-x})$$



DON, shared encoders, FiLM gates

Sine (SIREN)

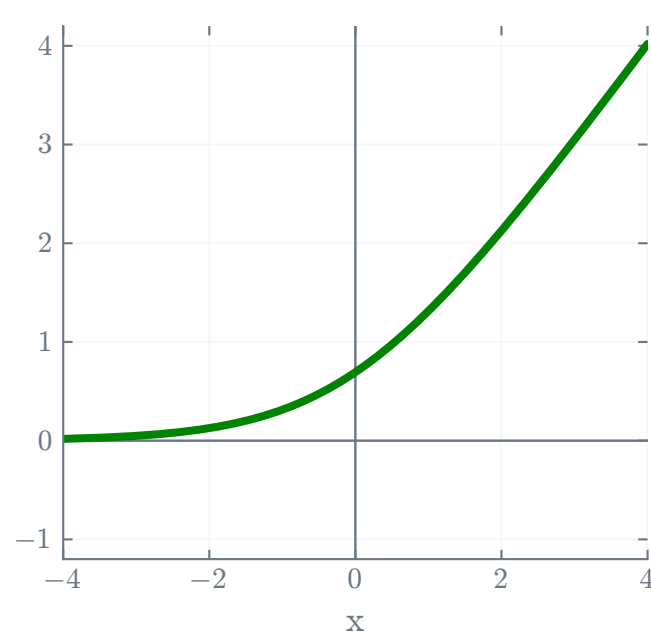
$$\sin(\omega_0 z), \omega_0 = 20$$



SIREN

Softplus

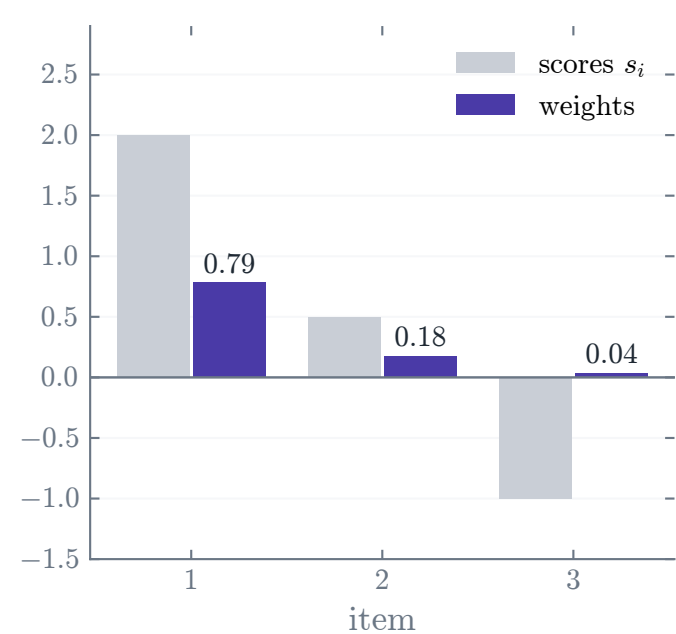
$$\ln(1 + e^x)$$



LNO (damping, frequency > 0)

Softmax

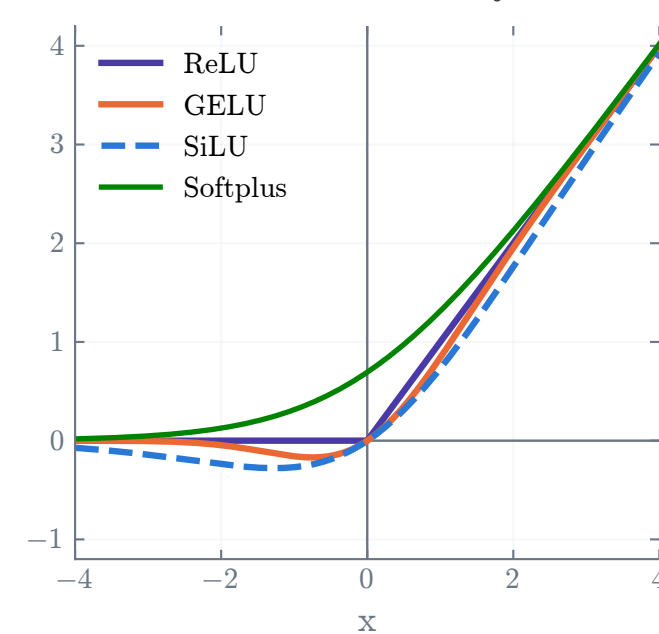
$$e^{s_i} / \sum_j e^{s_j}$$



attention in GNO, STO; DON term weights

Compared

the ReLU family



GELU and SiLU are smooth ReLUs